CHEM0011: Chemistry for Biologists

1. MODULE ORGANISER AND TEACHING TEAM

Module Organiser:	Dr. J L Kiappes
Contact details	chem001x@ucl.ac.uk [For Personal Issues only] – otherwise use forums
Laboratory Organiser:	UCLChemLab_14 –Dr M Blunt
Room number and email:	chem001x@ucl.ac.uk [For Personal Issues only] – lecturers will have student hours in their Moodle profile
Module Staff	*Lecturers*: Dr Anna Roffey (Structure and bonding) Dr Andrea Sella (On the Zeroes) Dr Sam Cassidy (Spectroscopy 1; Thermodynamics) Dr J L Kiappes (Spectroscopy 2) Dr Daren Caruana (Ions and Electrochemistry) Dr River Riley (Kinetics) Dr Matt Blunt (Lab_14) Dr Kreso Bucar (Functional Group Reactivity; Macromolecules) Dr Stephen Potts (Bio-inorganic Chemistry) *Others:* Postgraduate Mentors and Elves
Module Assessors:	As above

2. MODULE INFORMATION

Module Title:	Chemistry for Biologists
Module Code(s):	CHEM0011
Level of Study:	Level 4
Module Credits (ECTS)	30 credits (6 ECTS)
Expected Teaching Period:	Term 1 and Term 2
Normal prerequisite:	Chemistry A level
Board of Examiners:	Chemistry Undergraduate Board of Examiners

4. TEACHING AND LEARNING METHODS

Delivery of Material: The topics will be introduced through a blended delivery of in person lectures and workshops, an extensive set of online activities (including revision quizzes, discussion fora, online helpdesks, online videos, guided lessons) and a course textbook.

Workshops: Your learning is supported in each topic by revision workshops conducted in person. During the session lecturers will go over problem sets, past exams or cover topics as requested by students and provide worked examples to problem sets. Preparation for these workshops is expected and essential. You will be expected to submit a couple of worked answers in advance to Moodle for peer assessment, and to provide feedback to your peers.

Self-study: In addition to timetabled hours it is expected that you engage in self-study in order to master the material. You will have extensive resources available from workshop questions, revision quizzes on Moodle, past exam questions, and questions in the textbook. We also recommend setting up study groups so that students can revise together, testing each other on course material.

Practical work: The practical component of this course is 'Lab_14', in which you are provided with a boxed chemistry set to carry out a range of open-ended exercises and measurements to improve your skills in measurement, problem-solving, data analysis, and cooking. As practical skills are an essential aspect of this programme, engagement is monitored and contributes to your final grade.

5. TEACHING AND LEARNING PATTERN

Activity	Hours
Lectures	40
Workshops	20
Hands-on practicals	30
Self-Directed Learning	210
TOTAL HOURS	**300**

6. ASSESSMENT METHODS AND DEADLINES

The module will be assessed through an online exam (65%) in the main examination period (May); submission/upload of practicals (25%), submission and peer marking of pre-workshop questions (10%).

7. ASSESSMENT PATTERN

Assessment Component	*Type*	*Weighting*
Unseen online exam	Online Exam	65%
Submission and peer assessment of workshop questions	Upload and peer assessment	10%
Practical	Completion by submission of results	25%

Module Pass Mark	40%

8. MODULE OUTLINE

The aim of this module is to provide students with the key foundations in chemistry and the numeracy skills needed to develop an understanding of biological systems at a molecular level.

9. MODULE AIMS

At the end of the module students should be able to:

1. Understand the principles underpinning structure and bonding in ions and molecules of biological interest. 2. Understand the basic principles of intermolecular forces that lead to membrane self-assembly and protein folding.

3. Have an understanding of how molecules interact with light, including absorption, fluorescence and NMR spectroscopy.
4. Understand equilibrium and the factors which determine it (enthalpy, entropy and the Gibbs energy)
5. Understand the links between concentration and oxidation/reduction processes.
6. Understand pH and its measurement.
7. Formulate and to understand the factors influencing the rates of chemical reaction.
8. Use key transformations of major organic functional groups
9. Recognise key biological macromolecules, their structure and chemistry
10. Understand the role played by metal ions in biological systems.
11. Be able to out measurements and basic chemical procedures using substances that can be bought in the supermarket or corner shop, using using simple instruments like a multimeter, digital thermometer, pH meter and mobile phone.
12. Be familiar with order of magnitude estimates/calculations.

10. MAPPING OF MODULE AIMS TO ASSESSMENTS

Aims 1-12 will be assessed through the unseen exam and Moodle coursework tests.

11. LINKS TO OTHER MODULES

Students expecting to take higher level modules in chemistry and in biochemistry/molecular biology, pharmacology or neuroscience should take this module as it provides the foundation in molecular science required for fuller appreciation of more advanced concepts.

12. MODULE EVALUATION

The Module will be evaluated using the standard SEQ on Moodle as well as information gained from student representatives at the relevant Staff-Student Consultative committees.

13. MODULE SYLLABUS

OTZ – ON THE ZEROES

Three interactive workshops devoted to order of magnitude calculations and estimation designed to help with critical thinking.

A: Structure, Bonding, and Self-Assembly of Biological Structures

- Electron configuration of atoms, Pauli principle, Hund's rule. Covalent bonding in simple molecules – principally made up of C, O, H, N, S, P – using electron-sharing arguments (Lewis structures). Single and multiple bonds. Bond order.
- Geometry of covalently bonded molecules based on valence shell electron pair repulsion (VSEPR). Orbital overlap and hybridisation models for bonding (Valence Bond Theory). Sigma and pi bonds, rationalised using diagrams for orbital overlap.
- Basic Molecular Orbital Theory, as an alternative representation of bonding in molecules. Electronegativity, and its relation to bond polarity, and ionic vs. covalent bonding.
- The role of non-covalent interactions between molecules in determining the properties of condensed-phase and biological materials. The interaction between charges, dipoles, and induced dipoles in varying combinations. How molecular liquids behave as solvents, in particular their differing polarity.
- The properties of water as nature's solvent, in particular the dissociation and hydration of ionic species in water. The hydrogen bonding within bulk water, between water and solute molecules,

and within and between other molecules. The origin of hydrophobic interactions which cause the aggregation of non-polar moieties in water. Solvation of non-polar molecules.

- The concept of self-assembly, building from the non-covalent interactions studied previously.

B: Seeing Biomolecules – Spectroscopy and Quantification

- The nature of light and how it interacts with molecules
- Ground and Excited states
- UV-visible spectroscopy and electronic structure
- UV-vis absorption spectra and the Beer-Lambert Law
- Fluorescence of organic molecules
- Nuclear Magnetic Resonance (NMR) spectroscopy: basic concepts
- Proton (^{1}H) NMR spectra: shielding and chemical shifts
- High resolution ^{1}H NMR spectra: coupling patterns
- Interpreting NMR spectra of organic molecules

C: Chemical Equilibrium and Chemical Thermodynamics

- Definition of energy, kinds of energy, units,
- Energy consumption and power output – comparisons across different organisms
- The First Law – Conservation of Energy
- Heat and Work as mechanisms of energy transfer – Temperature difference as the driver of heat.
- The Second Law – The Quality of Energy
- Spontaneous vs Irreversible change – Free Energy as a criterion.
- The idea of ΔG ATP as fundamental biological energy unit.
- Simple introduction to Boltzmann's ideas
- Excursion into energy flows. Dissipative structures or Protein Folding

D: Concentrations, pH and Equilbria

- Revision of acids and bases. Strong and weak acids. Multiple ionisation.
- Buffers. Buffering capacity.
- Titration curves. Indicators.
- Revision of electrochemistry. Galvanic cells.
- Standard potentials and Free Energy.
- Chemical potential and extent of reaction.
- Equilibria in Redox reactions.
- Nernst equation. Application of the Nernst equation.
- Membrane potentials.
- Reference electrodes and applications in biology.

E: The Speed of Life – Chemical and Biological Kinetics

- Nature and scope of kinetics.
- Meaning and definition of rate of reaction.
- Factors influencing rate of reaction.
- Concept of order of reaction. Examples of reactions with simple orders. Pseudo-order reactions. Order and stoichiometry. Integrated rate equations for first and second order processes.
- Experimental methods in kinetics. Determination of reaction order and rate coefficients. Kinetics and reaction mechanism. Concepts of elementary processes and molecularity. The steady state approximation.
- Enzyme kinetics as an example of the application of the steady state approximation. The Michaelis-Menten equation for reactions involving one intermediate.
- Diffusion and what controls its rate
- Effect of temperature on reaction rate – the Arrhenius equation. Theories of reaction rates: collision theory and transition state theory. Catalysts.

- An excursion into biological pattern formation.

H: The Inorganic Chemistry of Life

- The Periodic Table and the elements of life
- Metal ions – hard and soft – polarizability and polarizing power
- Ligands and how they bind to metals
- Factors affecting binding strength – the stability constant
- Metal binding sites – selectivity
- The role of metal ions in biological systems – examples drawn from iron chemistry – hydrogenase, ferredoxin, haem proteins etc.

F: The Bricks of Biological Architecture – Functional Groups and Reactivity

Preparations and transformations of functional groups of relevance to biological applications, e.g.:

- Oxidation and reduction
- Substitution reactions
- Addition reactions
- Elimination reactions
- Formation of ethers, acetals and aldols

G: Macromolecules – the Spokes and Wheels of the Cell

The fundamental chemistry of proteins starting from structures and behaviour of carboxylic acids their reaction products:

- Carboxylic acids – factors determining pK_a
- Esters and fats
- Amides
- Amino acids and peptides
- Electrophoresis and isoelectric focussing.

14. RECOMMENDED READING LIST

Recommended Text

- Burrows, Holman, Parsons, Pilling, Price "**Chemistry^3**", Oxford UP plus additional readings from other texts in a custom bundle.

NB: This custom bundle will only be available from Waterstones, Gower Street. Copies are also available from the DMS Watson Library and on Biblius

Second hand copies of the text may also be available – look at bulletin boards round college. We will attempt to set up a clearinghouse for students to make this easier.

Alternative Chemistry Texts

There are many books in the DMS Watson library that are highly suitable.

- McMurry and Fay, "Chemistry", Pearson, any edition.
- Kotz and Purcell/Treichel, "Chemistry and Chemical Reactivity", Cengage, any edition.
- Brady and Holum, "Chemistry – The Study of Matter and its Changes", Wiley, any edition.
- Raymond Chang, "Physical Chemistry for the Biosciences", University Science Books, 2005. • Raymond Chang, "Physical Chemistry for the Chemical and Biological Sciences", University Science Books, 2000.
- P W Atkins, "The Elements of Physical Chemistry", Oxford University Press, any edition.

Advanced reading

- Phillips, Kondev, Theriot, "The Physical Biology of the Cell", Garland Science.
- Nelson, "Biological Physics", Freeman.
- Haynie "Biological Thermodynamics" Cambridge UP.

Reading for Rainy Days

- P Levi, "The Periodic Table", many editions • P Ball, "How Life Works ", U. Chicago, 2024.
- P Ball, "Nature's Patterns: A Tapestry in Three Parts", Oxford UP 2009 (3 vols. Shapes, Flow, Branches)
- H McGee, "On Food and Cooking – Science and Lore of the Kitchen", Prentice Hall, any edition.

15. LABORATORY COURSE (Lab 14)

You will be issued with a kit with which you will do practicals at home or where you are living. The practical tasks will require ingenuity and improvisation and are designed to develop your scientific thinking rather than arriving at specific "correct" results. The practicals will give you the opportunity to measure and observe a wide range of phenomena, some related to course material and some going well beyond it.

Practicals are supported by a separate website. The practical course will have a minimum completion requirement but you are encouraged to participate in as many activities across the year as you can.

Organisation:

The practicals will run week by week in your own time with an expected completion requirement by the weekend.

Helpdesks will be available through both physically and online using Zoom. There is also an online forum with mentors and academic staff available to answer questions.